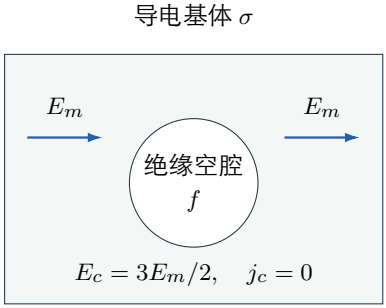
M4-021b　空腔场怎样进入端口电阻



体积分数 $f \ll 1$；形状不按比例

$$\begin{aligned}\overline{E} &= (1-f)E_m + fE_c \\ &= (1+f/2)E_m\end{aligned}$$

$$\overline{j} = (1-f)\sigma E_m$$

孔内电场不为零，传导电流为零

$$\begin{aligned}\sigma_{\text{eff}} &= \frac{\overline{j}}{\overline{E}} = \sigma\frac{1-f}{1+f/2} \\ &= \sigma(1-3f/2) + O(f^2)\end{aligned}$$

固定总电流时，电导率降低使端电压升高；不能仅按缺失的横截面积估算。